Examiner reports

Q1.

Throughout this question there seemed to be two categories of student: those who were familiar with this sort of practical exercise, and had carried it out, and those who had perhaps only read about it. The former found this an easy source of 7 or 8 marks; the latter demonstrated through their answers that they did not really understand what was expected.

In part (a), the variety of detail in the acceptable answers showed the understanding of these students.

Those who had completed part (a) could generally supply the missing numbers in part (b), but not all realised the importance of 5 significant figures for $f(x + h)$ in this process.

Part (c) was surprisingly poorly answered, with many giving the more general answer, $1 - 2x$, rather than the specific −5. There were also many cases of 5 or 0.

Many students used the formula given in the formula booklet to make a successful start in part (d). The better ones adapted this to $f(3 + h)$ and went rapidly to the answer of −5. Others persisted with $f(x + h)$ but did not always continue to obtain −5. It was common for $\frac{h - 2xh - h^2}{h}$ to be reached but when the h was cancelled the single h on the top just vanished. This led to a final answer of $-2x$ and then −6. It was common then to see a correct answer to (c) altered to match this incorrect value.

Q2.

This optimisation question proved to be the most challenging question on the paper with over 60% of students scoring 0 marks. Many made no attempt at all. Those who attempted did not help their chances by misquoting basic formulae such as the circumference of a circle and the volume of a cylinder. The lack of structure clearly had a major impact which meant that few were able to make any significant progress. Most students were unable to visualise the problem and transfer the given information into the required two equations. Fewer still then reduced these down to an equation in one variable to differentiate. Some students incorrectly thought that the weld length was the surface area. In some cases students recognised what they needed to do, but could not generate a formula that they could work with.

This type of question is one of the most challenging demands of the new specification, and schools and colleges will need to take every opportunity to consolidate the skills required to attempt these questions.

Q3.

Differentiation from first principles is new to the A-level maths specification and while many students seemed well practiced in the routine of part (a), only the strongest students could correctly explain how the gradient of the chord led to A being a stationary point.

The most frequent mistake seen in part (a) was the incorrect manipulation of negatives in the expansion of $(-4 + h)^3$. Some attempts were too piecemeal and did not form coherent arguments.

Around half of the students completed part (b) successfully.

The most common mistake in part (b) was to simply $h = 0$, rather than explain that as h tends to zero the gradient of the chord approached the gradient of the tangent. The most successful students simply used correct limit notation to show that the gradient of the curve was zero at *A* and hence a stationary point.

Q4.

In (a), most students realised that the domain would take values $x > 1$ in this question. However, in order to gain full marks it was necessary to state the domain in a complete form involving the notation $x \in \mathbb{R}$, which most students failed to do.

In (b), a large number of students started to differentiate the printed result for $f'(x)$, rather than differentiate $f(x)$. Many then realised their mistake and restarted the question on a continuation sheet, with a good number scoring full marks.

There was a wide spread of marks across question (c). Most students understood that they needed $f''(x) = 0$, though a number instead used $f'(x) = 0$. Of those students who successfully solved the correct equation, very few scored full marks, as they either failed to correctly deal with the solution $x = 1$ or failed to test either side of their correct solution $x = 4$. Only a small proportion of students identified the correct interval in (d).

Q10.

(a) The differentiation was almost always correct.

(b) (i) Having only a single derivative to select meant that the success rate in finding the rate of change was greater than some years. Nevertheless, some had difficulty simplifying $\frac{1}{2} - 2$ without a calculator.

(ii) Most students made the correct deduction from their value in part (b)(i) but often the explanations were not clear enough to earn the mark. It was necessary to write something like “ $\frac{dy}{dt} < 0$, therefore the “height is decreasing”, making specific reference to the **negative** value of the derivative.

(c) (i) Once again, the second derivative was usually found correctly, but arithmetic errors sometimes arose when finding the value of $\frac{d^2y}{dt^2}$ when $t = 2$.

(ii) The interpretation of the sign of $\frac{d^2y}{dt^2}$, as indicating whether y had a minimum or maximum value, seemed to be well known.

Q11.

The different format of this question took many students by surprise. Many weaker students were confused about the processes required. Some even used integration in part (a) and gradients in part (b) or thought that they were required to find the normal and the tangent in the two parts.

(a) The majority realised the need to substitute $x = 1$ into the given expression for $\frac{dy}{dx}$, but even that was not always evaluated correctly. Quite a few differentiated the given expression and scored no marks at all. Although most went on to find the equation of the tangent in the required form, a substantial number found the normal instead. Others hedged their bets by finding the equations of both the tangent and the normal, with no indication as to which was which. It was disappointing to see that several students, who obtained a correct equation for the tangent in the form $y - 4 = 9(x - 1)$, were unable to simplify this correctly to the form $y = 9x - 5$.

(b) When integration was attempted, it was usually done correctly with just the odd mistake in signs or coefficients. However, many failed to include the constant of integration, " + c" , and could proceed no further. Many who did have the " + c" stopped at that point. Some substituted $x = 1$ into their expression for x and equated this to zero. This could not be given any credit because they did not have an expression of the form $y =$ …. and so were unable to substitute $y = 4$. Many of those who found the correct value of the arbitrary constant failed to write the final equation in the form $y = f(x)$.

Q12.

(a) (i) Almost all candidates attempted to find $p(-3)$ and most were successful in showing that $p(-3) = 0$. It was pleasing to see many candidates taking heed of previous reports and writing a concluding statement such as "so $x + 3$ is a factor". Those who simply used long division were ignoring the request to use the Factor Theorem and earned no marks.

(ii) The majority of candidates used inspection or long division and found the correct quadratic factor. A few using the latter method did not always write their final answer as the product of factors as requested.

(b) (i) Almost all found the first derivative correctly, though a few added "+ c".

(ii) Several candidates presented an inadequate solution here. It was necessary to both equate their answer to part (b)(i) to zero and divided by 4 in order to obtain the given equation.

(iii) Solutions to this part were very disappointing. Most candidates merely substituted $x = -3$ into the given equation and were, in effect, repeating part (a). Those who attempted to solve the quadratic equation or found the value of the discriminant did not always give a satisfactory explanation for the single stationary point of the curve. For example "the discriminant is negative therefore the only real root is $x = -3$'.

(iv) Once again the second derivative was found correctly by most, although its value was sometimes incorrect as a result of poor arithmetic, particularly in the evaluation of the term $12(-3)^2$. Unfortunately, quite a few candidates divided their expression for $\frac{dy}{dx}$ by 4 before differentiating again and wrote $\frac{d^2y}{dx^2}$ = $3x^2$ – 4, which was incorrect.

(v) The interpretation of the sign of the second derivative was well known. However not all candidates gave an adequate reason. A statement such as "92 > 0" or "$\frac{d^2y}{dx^2}$ is positive" was required; an answer such as "it is positive" or "the value is greater than 0", without specific reference to the value from part (iv), was not sufficient.

Q13.

In part (a), differentiation had been well drilled and most candidates earned all 4 marks for the two derivatives. Occasionally, the + 1 was omitted or an extra 5 or + x was included, and a few felt the need to add + c to both their answers.

In part (b), those candidates who substituted x = –1 into their expression for $\frac{dy}{dx}$ often obtained a gradient of 12. However, not all used this value: some took the negative reciprocal and then formed the equation of the normal. Some candidates did not use their expression for $\frac{dy}{dx}$ in order to find the gradient but erroneously used the gradient of *AB*, with value 2, instead.

Surprisingly, a very small minority earned full marks in part (c). Most candidates verified the nature of the stationary point by considering the sign of the second derivative. However only a handful recognised the need to verify that the point was stationary in the first place, by showing that the first derivative was equal to zero.

In part (d)(i), most candidates were able to integrate the expression, with only the weakest candidates not making any attempt. Poor notation was used, with many including the integral sign after integrating. However dealing with the substitution of –1 caused many errors. Manipulation of fractions caused problems too; it was disappointing to see many candidates with correct fractions being unable to combine these to give a value of 8 or equivalent. Many candidates did not find the actual value of the definite integral until part (d)(ii), and on this occasion full credit was given, but this may not be the case in future.

In part (d)(ii), most candidates calculated the area of the relevant triangle to be 2 and subtracted this from their answer to part (d)(i). Errors included writing the area of the triangle $\frac{4 \times 2}{2}$ as and careless arithmetic such as 8 – 2 = 4.

Q14.

In part (a)(i), some students merely doubled the given equation and then divided by 2; this did not convince examiners and earned no marks. Students were expected to show the separate areas of the faces of the cuboid; the minimum working required was sight of $2xy$ + $6xy$ as well as $6x^2$, before combining these to give $6x^2 + 8xy = 32$ and hence the printed result. Some proved the result by considering half the surface area and, provided enough explanation was given, this also scored full marks.

In part (a)(ii), the majority earned a single mark for writing down a correct expression for the volume in terms of x and y. The algebra required to rearrange the equation from part (a)(i) to make y or xy the subject proved too difficult for many and once again highlighted students' algebraic weaknesses as they tried to reach the printed answer but violated various algebraic laws on the way.

In part (b), whilst the term $12x$ was almost invariably differentiated correctly, the same could not be said of the second term in the expression for V, and $-27x^2$ was a common incorrect answer.

In part (c)(i), although most students realised the need to substitute $x = \frac{4}{3}$ into their expression for $\frac{dy}{dx}$, not all showed the required working to prove that $\frac{dy}{dx} = 0$. It was not sufficient to write $12 - \frac{27\left(\frac{4}{3}\right)^2}{4} = 12 - 12 = 0$; evidence of correct manipulation of the fractions was needed. Once again, many students did not write a concluding statement with regard to the zero gradient implying that there was a stationary point.

In part (c)(ii), by far the most common error when finding $\frac{d^2y}{dx^2}$ was the omission of the minus sign. The final mark was still available even if students made errors in finding the second derivative, provided the substitution of $x = \frac{4}{3}$ was done correctly and the appropriate conclusion was drawn regarding the nature of the stationary point.

Q15.

In part (a)(i), the idea of y being increasing when $\frac{dy}{dx} > 0$ seemed unfamiliar to many students, who seemed content to substitute a few numerical values for x into the expression for $\frac{dy}{dx}$ rather than tackling the three-line proof. Consequently, very few scored full marks on this part.

Many students ignored the inequality signs throughout part (a)(ii). It was pleasing to see many students attempting to factorise the quadratic, often successfully, rather than relying on the formula. Perhaps more students would be successful in solving quadratic inequalities if they were to draw a sign diagram or sketch a graph. There were many correct solutions, but students must realise that $x < 2$, $x > \frac{4}{3}$ is not the same as $\frac{4}{3} < x < 2$

In part (b)(i), most students attempted to evaluate $\frac{dy}{dx}$ when $x = 2$. Not all were able to cope with the evaluation of $-6\,(2)^2$, which they needed to do in order to convince examiners that $\frac{dy}{dx}$ was indeed equal to zero. Once more, some failed to write a concluding statement or referred to a stationary point rather than the fact that the tangent was parallel to the x-axis.

In part (b)(ii), it was pleasing to see some fully correct solutions. Most recognised that the equation of the tangent at P was $y = 3$, but it was disappointing to see a number of students write $y - 3 = 9(x - 2)$ followed by $y = x + 1$ as their tangent equation. In finding the equation of the normal at Q, some, having correctly found that $\frac{dy}{dx} = 10$ when $x = 3$, used the value 10 instead of $-\frac{1}{10}$ for the gradient of the normal. Arithmetic errors were very common by those who used $y = mx + c$ when finding the value of c; typically $y = \frac{x}{10} + \frac{7}{10}$ was seen instead of $y = \frac{x}{10} - \frac{13}{10}$. Curiously, quite a few students who had previously obtained $y = 3$ as the equation of the tangent, and consequently the correct y-coordinate of the point R, then substituted the correct value of x into their incorrect equation for the normal and found a different, and now incorrect value for the y-coordinate of R.

Q16.

In part (a)(i) of this calculus question, where students were asked to verify that a given expression was equal to zero when $x = 2$, a minority of students failed to give adequate written evidence, particular when the order of calculation was important; the minimum examiners accepted here was $3 \times 4 - \frac{4}{4} - 11 = 0$, although $12 - 1 - 11 = 0$ was more convincing. In part (a)(ii) the differentiation of $-\frac{4}{x^2}$ was a hurdle not always overcome by the weaker students. However, those who first re-wrote it as $-4x^{-2}$ usually went on to differentiate it correctly. Many students scored the mark for giving a correct reason in part (a)(iii). In part (b) a large number of the students did realise that they needed to integrate the given expression for the gradient to find the equation of the curve but a significant number of them did not include the arbitrary constant which is essential here, and only a minority knew how to find its value from the given information. Although only a minority, it was still disappointing to see more students than expected writing down the equation of the curve in the form $y - y_1 = m(x - x_1)$.

Q17.

Part (a) proved to be a good source of marks, although a significant number of students lost a mark through poor notation, such as $\cos y^2$ or no $\frac{dx}{dy}$

Although many earned full marks in part (b), many other proofs were inadequate. Students were expected to use the identity $\tan^2 y + 1 = \sec^2 y$ and to show the expansion of $(x - 1)^2$ clearly. Too many fudged their working, wrote $\tan y + 1 = \sec y$ or only proved the known trig identity.

The result in part (c) was easily derived from the previous two parts, but there were many blanks here and many confused statements about inverse functions which were often taken as $\frac{1}{\tan y}$. Students who used the formula book appropriately generally earned the mark.

There were many fully correct solutions to part (d)(i). However it was distressing at this stage to see some horrendous algebra, with terms inverted in ways that were invalid and worthless.

Only the more able students were able to make progress on the last two parts of the question. Many students differentiated their quadratic instead of the expression for $\frac{dy}{dx}$. Some students earned the method mark in the final part, but few completed the whole answer, as their final explanation often had an element missing.

Q18.

Most students understood the basics of what was required although a minority of others in part (a) just differentiated the expression and then substituted $-2 + h$ for x, for which no credit was awarded. The majority of students wrote down the correct expansion of $(-2 + h)^4$ but then some forgot to add $(-2 + h)$ before subtracting 14 and so ended up with a term in the form c / h in their expression which they then lost to reach an expression in the right form to match what was printed. In part (b) most students used '$h \to 0$' and generally gained credit but the small minority who just stated '$h = 0$', rather than considering any limits, did not score.

Q19.

In part (a) it was good to see almost everyone being able to differentiate correctly.

A few then went on to divide their expression for $\frac{dy}{dx}$ by 2 or 3 or 6, but this was not penalised at this stage.

In part (b) most candidates tackled the verification that P was a stationary point first and generally did so correctly, although some did not present accurate working, writing things such as $18 + 6 - 12 = 0$. Many stopped at this point, although a few tried listing various values of x with the corresponding value of $\frac{dy}{dx}$, and occasionally discovered the other value of x for which y was stationary. Many candidates equated $\frac{dy}{dx}$ to zero, but not all were then able to factorise the quadratic correctly. Some felt the need to use the quadratic equation formula and were not always successful in finding the correct solutions.

Part (c)(i) Those who had the correct expression for $\frac{dy}{dx}$ usually found the second derivative correctly. Many ignored the request to find the value of $\frac{d^2y}{dx^2}$ when $x = -1$, but this was sometimes generously credited in part (c)(ii), where it was pleasing to see most candidates being able to interpret the sign of the second derivative in order to describe the nature of the stationary point.

Q20.

In part (a) finding $\frac{dV}{dt}$ caused few problems apart from those who chose to multiply by 4 before

differentiating. Some will insist on adding $+c$ to their derivative despite advice given in previous reports.

In (b)(i) the concept of "rate of change" seemed to be better understood this time, helped by there being only one derivative to choose from; careless arithmetic when trying to simplify
0.75 – 3 spoiled many solutions.
Part (b)(ii) Most candidates realised that the volume was decreasing when t = 1 but many failed to give an adequate reason; most neglected to state the crucial fact that the rate of change was negative.

In part (c)(i) candidates were expected to solve an equation of the form $\frac{3}{4}t^2 - 3 = 0$ but most
chose to guess that t = 2 was the solution and attempted to verify this fact.
Part (c)(ii) Once again the explanation as to why the stationary value was a minimum value was inadequate, very often taking no account of the value of t found in the previous part of the

question. Some candidates who correctly found that the second derivative was $\frac{3t}{2}$ and

proceeded to solve the equation $\frac{3t}{2} = 0$ often obtained a value such as 6 and stated that this
positive solution indicated a minimum value.

Q21.

This was a very straightforward question for candidates who were well versed in the necessary techniques. Even when sign errors and other small slips occurred, it was usually possible to obtain high marks. Most candidates knew that in part (b)(ii) they should mention the fact that h tends to zero, though it cannot actually be equal to zero for the question to make sense.

Q22.

In part (a), almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

In part (b), those who substituted t = 2 into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt}$ = 0 is the condition for a stationary point. Some **assumed** that a stationary point occurred when t =

2, went straight to the test for maximum or minimum and only scored half the marks.

It was advisable to use the second derivative test here; those who considered values of $\frac{dy}{dt}$ on either side of t = 2 usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c)(i), the concept of 'rate of change' was not understood by many who failed to realise the need to substitute t = 1 into $\frac{dy}{dt}$. Some candidates wrongly substituted t = 1 into the initial expression for y or into their expression for $\frac{d^2y}{dt^2}$ and these candidates were unable to score any marks at all on this part. Even those who used $\frac{dy}{dt}$ sometimes made careless arithmetic errors when adding three numbers.

In part (c)(ii), it was not enough to simply write the word "increasing": some explanation about $\frac{dy}{dt}$ being positive was also required. Some candidates erroneously found the value of the second derivative when t = 1 or calculated the value of y on either side of t = 1.

Q23.

Part (a) was well answered with many candidates gaining full marks. Where part marks were earned, the most common error was to lose the factor of 4 in the numerator by incorrectly writing the derivatives of sin $4x$ and cos $4x$ as cos $4x$ and –sin $4x$.

Many candidates did not attempt part (b). Although fully correct responses were seen, this was the question part where most candidates scored very few if any marks. Where candidates split the term $4\tan^2 4x$ up into ($4\tan 4x$)($\tan 4x$) or similar expressions they seemed to have more success. In general, the use of the chain rule from whatever starting point was not coped with satisfactorily.

Q24.

Most candidates answered part (a)(i) correctly, although a surprising number seemed to think that $A \times 0 = A$.

Most candidates did part (a)(ii) convincingly, usually choosing to evaluate the exponential expression as 0.95 … and solving for A to demonstrate its value is 60 to two significant figures. Those who did not actually solve for A or did not show A as 59.9…were penalised.

Most candidates started part (b)(i) well, showing they had interpreted the information given correctly. However many could not manipulate their resulting In expression into the form required; many, either not knowing what was meant by "a and b are integers" or not taking notice of it, left their result in fractions. Many candidates 'lost' a minus sign during their working whilst some attempted to take the In of a negative number.

In part (b)(ii), a few candidates were able to derive the result clearly and efficiently; others managed it after some working that was somewhat difficult to follow whilst others made an error such as dropping a minus sign but still claimed they had obtained the result.

However, many candidates gave up after finding $\frac{dh}{dt}$. A few candidates attempted to integrate the given differential equation, some successfully recovering the given expression for A, which was both acceptable and impressive.

Although many candidates had part (b)(iii) correct, many errors were also seen, quite a common one being 4 × 2 = $\frac{1}{2}$. The other common error was to substitute h = 13, rather than $\frac{dh}{dt}$ = 13. It was notable that many candidates attempted part (b)(iii) before returning to try and sort out part (b)(ii), which was sensible.

Q25.

Part (a)(i) Usually after a few abortive attempts many candidates realised that they had to add together the areas of the various faces. Once they had the correct expression for the total surface area most candidates were able to obtain the printed result. There was clearly some fudging on the part of weaker candidates and they could earn little more than a single method mark.

Part (a)(ii) This was surprisingly one of the biggest discriminators on the paper with only the best candidates being able to obtain the correct expression for V. Trying to make y the subject of the equation from part (a)(i) caused problems for many who took this approach; others substituted for xy in the formula $V = 6x(xy)$ and were often more successful. Once again it was quite common to see totally incorrect expressions being miraculously transformed into the printed answer.

Part (b)(i) Most candidates scored full marks for this basic differentiation although it was not always clearly identified as $\frac{dV}{dx}$ in their working.

Part (b)(ii) Most substituted x = 2 into their expression for $\frac{dV}{dx}$ and found the value to be zero. It was then necessary to make a statement about the implication of there being a stationary value in order to score full marks.

Part (c) Some made a sign error when finding the second derivative, but the majority of candidates scored full marks in this part. Credit was given if the correct conclusion was drawn from the sign of their second derivative, provided no further arithmetic errors occurred.

Q26.

Most candidates coped with the negative fractional powers well, differentiated correctly and found an equation of a normal, although there were a few equations of tangents presented as final answers to part (b)(ii). A common arithmetical slip in part (b)(ii) was ‘

$\frac{1+\sqrt{1}}{1}$' incorrectly evaluated as 1. The expression for the second derivative was often correct in part (c)(i).

Part (c)(ii) was discriminating. The most popular approaches were to either put $x = 1$ in $\frac{d^2y}{dx^2}$, or to equate $\frac{d^2y}{dx^2}$ to zero and solve, for which no marks were scored.

A significant number of candidates did, however, earn some credit by making a more general comment on the sign of the second derivative but many lacked rigour and, in particular, assumed the second derivative has to be negative for a maximum, overlooking the possibility that the second derivative may be zero at a maximum.

Q27.

As has happened in past papers on MFP1, the expansion of the cube of a binomial expression involved some lengthy pieces of algebra for many candidates, though the correct answer was often legitimately obtained. Most candidates were then able to put all the necessary terms into the formula for the gradient of a straight line and obtain the required answer correctly. There was a good response to part (b), where many candidates stated correctly that h must **tend** to zero. Only rarely did they say, inappropriately, that it must be *equal* to zero.

Q28.

In part (a) almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

Those who substituted $t = 3$ into $\frac{dx}{dt}$ in part (b) did not always explain that $\frac{dx}{dt} = 0$ is the condition for a stationary point. Some assumed that a stationary point occurred when $t = 3$ and went straight to the test for maximum or minimum and only scored half the marks. It was advisable to use the second derivative test here; those who considered values of $\frac{dx}{dt}$ on either side of $t = 3$ usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c) the concept of “rate of change” was not understood by many. Approximately equal numbers of candidates substituted $t = 1$ into the expression for $\frac{dx}{dt}$ or $\frac{d^2x}{dt^2}$ and so only about half of the candidates were able to score any marks on this part. Those who used $\frac{dx}{dt}$ often made careless arithmetic errors when adding three numbers.

In part (d), as in part (c), candidates did not realise which expression to use and many wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a particular point.

Q29.

In part (a), most candidates used the product rule correctly, but at this stage $e^{2x} \neq 0$ was rarely seen, although many candidates obtained the correct quadratic equation. Factorisation, or use of the formula, was usually done well and many candidates obtained both of the correct solutions. Where candidates seemed unsure of what to do with the e^{2x}, many expressions involving the use of ln $e^{2x} = 2x$ were seen together with the formation of a cubic expression in x. A third value of $x = 0$ was also seen by candidates who thought this was the solution of $e^{2x} = 0$.

Part (b) was generally well answered by those candidates able to do the product rule in part (a). Some errors with the coefficients meant candidates lost accuracy marks.

In part (c), although many correct solutions were seen some candidates lost marks through incorrect evaluation of $\frac{d^2y}{dx^2}$.

Q30.

In part (a) almost everyone obtained the correct expression for $\frac{dy}{dx}$, although a few spoiled their solution by dividing each term by 5 or adding "+ c" to their answer.

In part (b) most candidates substituted $x = -2$ into their expression for $\frac{dy}{dx}$, but, in order to score full marks, it was necessary to show $(-2)^4$ written as 16 or to show that $\frac{dy}{dx} = 80 - 80 = 0$ and then to write an appropriate conclusion about P being a stationary point.

For part (c) many candidates simply wrote down an expression for $\frac{d^2y}{dx^2}$ in terms of x when answering part (i) and only evaluated the second derivative when determining the nature of the stationary point in part (ii). On this occasion full credit was given, but candidates need to realise what is meant by the demand to "find the value of" since this may be penalized in future examinations.

In part (d) some candidates failed to find the y-coordinate of P, which was necessary in order to find the equation of the tangent. It was pleasing to see most candidates using the value of $\frac{dy}{dx}$ when $x = 1$, but unfortunately many tried to find the equation of the normal instead of the tangent to the curve.

Q31.

Almost all candidates knew how to find the gradient of the line in part (a) of this question, but a distressingly large minority of them made a sign error in subtracting the y-values of A and B, which led to the introduction of an unwanted term $-\frac{6}{h}$ in the answer. In part (b) this should have caused difficulties, but almost invariably the candidates took this term as tending to zero as h tended to zero. Those candidates who wrote "h = 0" instead of letting h tend to zero lost a mark here, but it is pleasing to report that this error was comparatively rare.

Q32.

In part (a), most candidates were able to find the correct expression for $\frac{dy}{dx}$, although there were some who left + 5 in their answer or added +C.

In part (b), It had been expected that candidates would solve the equation $\frac{dy}{dx}$ = 0 and obtain the equation x^3 = 8 and hence deduce that x = 2. It seemed, however, that many were unable to formulate an appropriate equation, but merely spotted the correct answer: x = 2. This was not penalised on this occasion, provided that the candidate stated clearly that the x-coordinate of M was equal to 2.

In part (c)(i), the expression for $\frac{d^2y}{dx^2}$ was usually correct.

In part (c)(ii), although the method was left open, most candidates found the value of the second derivative when x = 2 and correctly concluded that M was a minimum point.

In part (c)(iii), some candidates were not aware of the need to find the value of $\frac{dy}{dx}$ when x = 0 in order to ascertain whether the curve was increasing or decreasing at that point.

Q33.

Most candidates performed well in this question.

Part (a)(i) seemed to present a stiff challenge to their algebraic skill, though they often came through the challenge successfully after several lines of working. Part (a)(ii) again proved harder than the examiners had intended, some candidates having to struggle to evaluate the y-coordinate of the point A. Again the outcome was usually successful, though a substantial minority of candidates used differentiation of the answer to the previous part. In part (a)(iii) it was pleasing to see that a very good proportion of the candidates correctly mentioned h 'tending to' zero rather than 'being equal to' zero, which was not allowed, though the correct value of the gradient could be obtained by this method and one mark was awarded for this.

The responses to part (b) suggested that most candidates were familiar with the Newton-Raphson method and were able to apply it correctly, but that they lacked an understanding of the geometry underlying the method, so that they failed to draw a tangent at the point A on the insert as required, or failed to indicate correctly the relationship between this tangent and the x-axis.

Q34.

Some candidates did not attempt part (a). However, those who could square 9 – 3x usually earned the marks. A surprising number substituted for both x and y and most were unable to complete the solution successfully.

In part (b)(i), the differentiation was carried out very well by most candidates. Errors in the value of k occurred due to taking out a common factor either before or after differentiating, so values of k such as 3 and 9 were common. A few weak candidates did not recognise that they had to differentiate V here, and generally were unable to proceed through the rest of the question.

In part (b)(ii), the factorisation was usually carried out successfully, and candidates who had not found the correct value of k were still able to earn these marks.

The most common error in part (c) was differentiating $x^2 - 4x + 3$, rather than $27(x^2 - 4x + 3)$, and this approach earned a method mark only.

Credit was given in part (d)(i) for substituting their values of x into their second derivative and so most candidates earned this mark, provided they had two values of x.

In part (d)(ii), most realised that $x = 1$ needed to be selected since $\frac{d^2y}{dx^2}$ was negative when $x = 1$.

In part (d)(iii), not all substituted $x = 1$ into the equation for V. Those who did often made arithmetic errors. Some gave the value of y when $x = 1$ and others gave the answer as –54, the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

Q35.

Part (a) Candidates did not seem confident at working on this kind of problem and algebraic weaknesses were evident. Many worked backwards from the result in part (a)(i) and did not always convince the examiner that they were considering the surface area of four faces and the base. The inability of most candidates to rearrange the formula to make h the subject in part (a)(ii) was alarming. Consequently few, without considerable fudging, could establish the printed formula for the volume. Part (b) Basic differentiation is well understood and most candidates found $\frac{dV}{dx}$ correctly. Some tried to substitute $x = 3$ into the expression for V in order to show there was a stationary point, but usually this part was answered well.

Part (c) It was not uncommon to see the second derivative as 4x even though the first derivative was correct. A generous follow through was given here provided candidates could interpret the value of their second derivative.